8.6.11 Lesson Review

Date: 4/4/2026, 10:52:30 AM

Time Spent: 05:18

Score: 87%

Passing Score: 80%

Question 1

Correct

What is the primary function of traffic shapers?

- To increase the speed of packet delivery
- To delay certain packet types based on their content ✓ Correct
- To categorize protocols into different service levels
- To monitor traffic conditions and generate reports

Explanation

The correct answer is to delay certain packet types based on their content. Traffic shapers, or bandwidth shapers, delay certain packet types based on their content to ensure that other packets have a higher priority. This helps reduce latency for critical applications by managing the flow of traffic more effectively.

Traffic shapers do not directly increase the speed of packet delivery; they manage traffic flow to prioritize certain packets.

Categorizing protocols into different service levels is a function of Class of Service (CoS), not traffic shaping.

Monitoring traffic conditions and generating reports is more aligned with the management plane's functions, not specifically traffic shaping.

Related Content

resources\questions\q_traffic_shape_traffic_shaper_role_n09.question.xml

Question 2 ✕ Incorrect

What is a potential consequence of using traffic policing devices instead of traffic shapers?

- (●) Increased latency for non-critical applications ✕ Incorrect
- ( ) More efficient use of available bandwidth
- ( ) Fewer lost packets during periods of congestion
- ( ) Under-utilization of bandwidth during idle periods ✓ Correct

Explanation

Traffic policing devices may simply drop packets once a traffic threshold is reached, which can lead to under-utilization of bandwidth during periods when the network is relatively idle. This contrasts with traffic shapers, which aim to use bandwidth more consistently by storing and managing packet delivery.

Increased latency for non-critical applications is not a direct consequence of using traffic policing devices; it's more related to how traffic priorities are managed.

Traffic policing devices may not lead to more efficient use of available bandwidth since they can result in packet loss and under-utilization during idle periods.

Fewer lost packets during periods of congestion is more characteristic of traffic shaping, which manages packet delivery to minimize loss, rather than traffic policing.

Related Content

resources\questions\q_traffic_shape_traffic_policing_consequence_n09.question.xml

Question 3 Correct

A network administrator is tasked with improving the performance of a company's VoIP (Voice over Internet Protocol) system, which has been experiencing poor audio quality during peak business hours.

The network is also used for email, web browsing, and file transfers. The administrator decides to implement a solution to prioritize VoIP traffic over other types of traffic.

Which of the following solutions would be MOST effective in achieving the desired improvement in VoIP performance?

- Implementing a First-In, First-Out (FIFO) queuing discipline across the network.
- Configuring a Random Early Detection (RED) algorithm on all network switches.
- Deploying traffic shaping to prioritize VoIP packets. ✓ Correct
- Increasing the bandwidth of the company's internet connection.

Explanation

Traffic shaping is the most effective solution for prioritizing VoIP traffic over other types of traffic. By delaying certain packet types based on their content, traffic shaping ensures that VoIP packets, which are sensitive to latency, are given higher priority. This can significantly improve the audio quality of VoIP calls during peak hours by reducing latency and jitter for these packets.

FIFO does not prioritize traffic based on content or type, leading to no improvement in VoIP performance during congestion.

RED is used to manage buffer overflow by preemptively dropping packets, which does not inherently prioritize VoIP traffic.

Increasing the bandwidth might alleviate some issues caused by congestion but does not address the need for prioritizing VoIP traffic over less time-sensitive traffic types.

Related Content

- 7.3.4 VoIP Protocols
- 8.6.1 Common Performance Issues
- 8.6.5 Bandwidth Management
- 8.6.6 Traffic Shaping

resources\questions\q_traffic_shape_scenario_01_n09.question.xml

Question 4 Correct

What advantage do website performance checkers offer for testing site response times?

- They can physically repair websites.
- They test from the perspective of customers in different countries. ✓ Correct
- They increase website traffic automatically.
- They design websites.

Explanation

Website performance checkers allow users to test a site's response times from various global locations, simulating the experience of international customers. This is crucial for understanding and optimizing the performance of a website for users worldwide.

Website performance checkers are tools for testing, not for physically repairing websites.

These tools measure performance; they do not automatically increase website traffic.

The purpose of website performance checkers is to test site response times, not to design websites.

Related Content

- 8.4.1 Network Device Logs
- 8.5.1 Packet Capture
- 8.6.1 Common Performance Issues
- 8.6.2 Interface Statistics
- 8.6.3 Flow Data
- 8.6.4 Traffic Testing Tools
- 8.6.5 Bandwidth Management

resources\questions\q_traffic_analyze_website_checkers_n09.question.xml

Question 5 Correct

What do "top talkers" and "top listeners" refer to in network analysis?

- Top talkers are devices with the highest error rates in the network, while top listeners are devices with the highest packet loss rates.
- Top talkers are the most secure connections in a network, and top listeners are the least secure connections.
- Top talkers are interfaces generating the most outgoing traffic, while top listeners are the interfaces receiving the most incoming traffic. ✓ Correct
- Top talkers are the fastest network connections, while top listeners are the slowest network connections.

Explanation

In network analysis, "top talkers" and "top listeners" are terms used to identify network interfaces based on their traffic volume. Top talkers are those interfaces that send out the most data, indicating high outgoing traffic. This could be due to applications or devices that are heavily transmitting data across the network. On the other hand, top listeners are interfaces that receive the most data, indicating high incoming traffic. Identifying these interfaces helps network administrators understand traffic flow, pinpoint potential bottlenecks, and optimize network performance.

"Top talkers" and "top listeners" do not refer to error rates or packet loss rates. Instead, these terms specifically relate to the volume of traffic being sent or received by network interfaces. Error rates and packet loss are different metrics used in network analysis to assess the quality of connections and the reliability of data transmission, not the volume of traffic.

The concepts of "top talkers" and "top listeners" have nothing to do with security levels of connections. These terms are used to quantify traffic volume, not to evaluate the security or vulnerability of network connections. Security assessments in a network involve analyzing encryption standards, authentication protocols, and other security measures, not traffic volume.

"Top talkers" and "top listeners" refer to the amount of data being transmitted or received, not the speed of the connections. The speed of a network connection is determined by its bandwidth and latency, among other factors, and is a separate consideration from the volume of traffic that a connection handles. Identifying top talkers and top listeners helps in understanding traffic distribution and potential bottlenecks but does not directly measure the speed of network connections.

Related Content

- 8.4.1 Network Device Logs
- 8.5.1 Packet Capture

8.6.1 Common Performance Issues

8.6.2 Interface Statistics

8.6.3 Flow Data

8.6.4 Traffic Testing Tools

8.6.5 Bandwidth Management

resources\questions\q_traffic_analyze_top_talkers_listeners_n09.question.xml

Question 6

Correct

What is packet loss?

- The encryption of data packets for security purposes
- The intentional slowing down of data transmission
- The discarding of data packets in a network ✓ Correct
- The duplication of data packets to ensure delivery

Explanation

The correct answer is the discarding of data packets in a network. Packet loss occurs when one or more data packets being transmitted across a network fail to reach their destination. This can be due to various reasons, such as network congestion, faulty hardware, or signal degradation, leading to the loss of data packets.

Encrypting data packets is a method of securing data, not a form of packet loss.

Intentionally slowing down data transmission is related to traffic management, not packet loss.

Duplication of data packets is a method to ensure delivery in some protocols, not a description of packet loss.

Related Content

8.6.1 Common Performance Issues

8.6.2 Interface Statistics

8.6.3 Flow Data

12.4.2 Insufficient Wireless Coverage Issues

resources\questions\q_qos_metrics_packet_loss_description_n09.question.xml

Question 7 Correct

What are latency and jitter in the context of network performance?

- Types of network security protocols
- Measures of network speed and efficiency
- Problems of timing and sequence of packet delivery ✓ Correct
- Techniques for optimizing bandwidth usage

Explanation

Latency refers to the delay before a transfer of data begins following an instruction for its transfer, while jitter is the variation in the delay of received packets. Both are crucial in determining the quality of real-time applications like VoIP and video streaming, as they affect the smoothness and quality of the transmission.

Latency and jitter are not security protocols but are related to the performance of data transmission.

While they can affect network speed and efficiency, latency and jitter specifically refer to timing and sequence issues in packet delivery.

These are not techniques for optimizing bandwidth usage but are issues that can negatively impact the quality of network services.

Related Content

8.6.1 Common Performance Issues

resources\questions\q_qos_metrics_latency_and_jitter_description_n09.question.xml

Question 8 Correct

What type of data transfer is described as bursty?

- Real-time video streaming
- Voice over IP (VoIP)
- File transfer ✓ Correct
- Live audio broadcasting

Explanation

Bursty data transfer refers to the transmission of data where large amounts of data are sent in bursts, with variable intervals between them. This is typical of file transfers, where the focus is on moving data as quickly as possible without strict requirements on the order or timing of packet delivery.

Real-time video streaming requires consistent data flow and is sensitive to delays, making it not bursty.

VoIP is a real-time service that requires consistent and timely delivery of packets, which is opposite to the nature of bursty data transfer.

Live audio broadcasting, like other real-time services, requires a steady stream of data and is not considered bursty.

Related Content

- 7.3.4 VoIP Protocols
- 8.6.1 Common Performance Issues
- 8.6.5 Bandwidth Management
- 8.6.6 Traffic Shaping

resources\questions\q_qos_metrics_bursty_data_transfer_n09.question.xml

Question 9 Correct

Why is collecting just the packet metadata preferred over recording each frame?

- It increases the network speed.
- It imposes a heavy processing overhead.
- It reduces the bandwidth required by the sniffer. ✓ Correct
- It enhances the security of the network.

Explanation

Collecting only packet metadata (such as source and destination addresses, protocol type) instead of entire packet payloads significantly reduces the amount of data that needs to be processed and transmitted, thereby reducing the bandwidth required for sniffing activities.

While reducing bandwidth requirements for sniffing activities can help maintain network performance, the primary benefit is the reduced data processing and storage needs, not directly increasing network speed.

Imposing a heavy processing overhead is actually the opposite; recording each frame imposes a heavy processing overhead, which is why collecting just the metadata is preferred.

While metadata collection can be part of security monitoring strategies, the primary reason for choosing it over full packet capture is to reduce bandwidth and processing requirements, not directly to enhance security.

Related Content

- 8.4.1 Network Device Logs
- 8.5.1 Packet Capture
- 8.6.1 Common Performance Issues
- 8.6.2 Interface Statistics
- 8.6.3 Flow Data
- 8.6.4 Traffic Testing Tools
- 8.6.5 Bandwidth Management

resources\questions\q_netflow_packet_metatdata_n09.question.xml

Question 10 Correct

What is the primary purpose of monitoring individual interface statistics in a network?

- To increase the network speed
- To diagnose performance issues ✓ Correct
- To reduce the cost of network maintenance
- To enhance the security of network communications

Explanation

The primary purpose is to diagnose performance issues. Monitoring interface statistics is essential for identifying and diagnosing network performance issues. By analyzing metrics such as bandwidth usage, packet loss, and errors, network administrators can pinpoint problems and take corrective actions to ensure optimal network performance.

While monitoring can help identify bottlenecks that, once resolved, may improve network speed, the primary purpose is diagnostics, not direct speed enhancement.

While efficient network operation can indirectly lead to cost savings, the primary goal of monitoring interface statistics is to diagnose performance issues, not to reduce maintenance costs directly.

Security is a critical aspect of network management, but the primary purpose of monitoring interface statistics is to diagnose performance issues, not to directly enhance security.

Related Content

- 6.1.3 TCP Handshake and Teardown
- 8.2.5 Performance Monitoring
- 8.4.5 Log Reviews
- 8.6.3 Flow Data
- 8.6.4 Traffic Testing Tools

resources\questions\q_netflow_interface_statistics_monitor_n09.question.xml

Question 11 ✕ Incorrect

What role does a NetFlow exporter play in a network?

- It aggregates flows from multiple collectors.
- It analyzes and reports flow information. ✕ Incorrect
- It defines cache flows. ✓ Correct
- It increases the bandwidth of the network.

Explanation

A NetFlow exporter is a component configured on network devices like routers and switches. It is responsible for defining traffic flows based on specific characteristics and caching this data for transmission to a NetFlow collector.

Aggregating flows from multiple collectors is the role of a NetFlow collector, not an exporter.

Analysis and reporting are functions of a NetFlow analyzer, not an exporter.

A NetFlow exporter's role is to define and cache traffic flows, not to increase network bandwidth.

Related Content

- 8.4.1 Network Device Logs
- 8.5.1 Packet Capture
- 8.6.1 Common Performance Issues
- 8.6.2 Interface Statistics
- 8.6.3 Flow Data
- 8.6.4 Traffic Testing Tools
- 8.6.5 Bandwidth Management

resources\questions\q_netflow_exporter_role_n09.question.xml

Question 12 Correct

What is the significance of monitoring queue length in network performance?

- It indicates the total number of users in the network.
- It helps determine whether the link is a bottleneck. ✓ Correct
- It shows the financial cost of data transfer.
- It measures the physical condition of the network interface.

Explanation

Monitoring queue length can reveal if data is consistently queued for transmission, indicating a potential bottleneck in the network.

Queue length relates to data packets waiting to be transmitted, not user count.

Queue length concerns data flow, not costs.

It's a measure of data traffic management, not the physical state of hardware.

Related Content

- 8.6.1 Common Performance Issues
- 8.6.2 Interface Statistics
- 8.6.3 Flow Data
- 8.6.5 Bandwidth Management
- 8.6.6 Traffic Shaping

resources\questions\q_int_stats_queue_length_monitor_n09.question.xml

Question 13

Correct

What is considered an acceptable error rate in general terms?

- (●) Under 1 percent ✓ Correct
- ( ) Exactly 5 percent
- ( ) Under 10 percent
- ( ) Over 15 percent

Explanation

A general guideline is that error rates should be under 1 percent. Higher rates may indicate issues that need addressing.

5 percent is significantly higher than the acceptable threshold.

10 percent is far above the acceptable error rate, indicating serious issues.

15 percent would suggest critical problems with the network.

Related Content

- 8.6.1 Common Performance Issues
- 8.6.2 Interface Statistics
- 8.6.3 Flow Data
- 12.4.2 Insufficient Wireless Coverage Issues

resources\questions\q_int_stats_error_rate_acceptable_n09.question.xml

Question 14 Correct

What is the primary purpose of using Quality of Service (QoS) mechanisms in a network?

- To decrease network security
- To increase the number of devices that can connect to the network
- To prioritize certain types of traffic over others ✓ Correct
- To reduce the cost of network infrastructure

Explanation

The primary purpose of QoS mechanisms is to manage bandwidth and prioritize certain types of traffic, such as VoIP or streaming video, over less critical applications. This ensures that important applications receive the necessary bandwidth and performance levels, even during times of high network congestion.

QoS mechanisms are designed to manage traffic and bandwidth, not to decrease network security.

Increasing the number of devices that can connect to the network is more related to network capacity and infrastructure than to QoS.

While effective use of QoS can optimize the use of existing network infrastructure, its primary purpose is not to reduce costs but to prioritize traffic.

Related Content

- 7.3.4 VoIP Protocols
- 8.6.1 Common Performance Issues
- 8.6.5 Bandwidth Management
- 8.6.6 Traffic Shaping

resources\questions\q_band_manage_qos_purpose_n09.question.xml

Question 15 Correct

Which of the following is NOT a traffic class defined by DiffServ?

- Best Effort
- Assured Forwarding
- Expedited Forwarding
- Maximum Throughput ✓ Correct

Explanation

The correct answer is maximum throughput. DiffServ traffic classes include Best Effort, Assured Forwarding, and Expedited Forwarding. Maximum Throughput is not a defined traffic class within the DiffServ framework, which focuses on prioritizing packets based on their classification.

Best Effort is a basic DiffServ traffic class where packets are delivered based on available capacity without special priority.

Assured Forwarding is a DiffServ traffic class that provides guaranteed delivery under specified conditions.

Expedited Forwarding is a high-priority DiffServ traffic class designed for time-sensitive data.

Related Content

- 8.6.1 Common Performance Issues
- 8.6.2 Interface Statistics
- 8.6.3 Flow Data
- 8.6.4 Traffic Testing Tools
- 8.6.5 Bandwidth Management
- 8.6.6 Traffic Shaping
- 8.6.7 Lab: Configure QoS
- 12.4.1 Wireless Performance Assessment
- 12.4.6 Overcapacity Issues

resources\questions\q_band_manage_maximum_throughput_n09.question.xml

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